

Dual-Targeted Lactoferrin Shell-Oily Core Nanocapsules for Synergistic Targeted/Herbal Therapy of Hepatocellular Carcinoma

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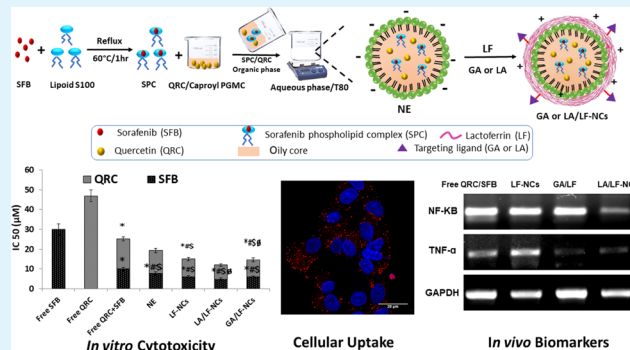
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Supporting Information

ABSTRACT: Herein, both strategies of synergistic drug combination together with dual active tumor targeting were combined for effective therapy of hepatocellular carcinoma (HCC). Therefore, based on the tumor sensitizing action, the herbal quercetin (QRC) was co-delivered with the targeted therapeutic drug sorafenib (SFB), preformulated as phospholipid complex, via protein shell-oily core nanocapsules (NCs). Inspired by the targeting action of lactoferrin (LF) via binding to LF receptors overexpressed by HCC cells, LF shell was electrostatically deposited onto the drug-loaded oily core to elaborate LF shell-oily core NCs. For dual tumor targeting, lactobionic acid (LA) or glycyrrhetic acid (GA) was individually coupled to LF shell for binding to asialoglycoprotein and GA receptors on liver cancer cells, respectively. Compared to LF and GA/LF NCs, the dual-targeted LA/LF-NCs showed higher internalization into HepG2 cells with 2-fold reduction in half-maximal inhibitory concentration compared to free combination therapy after 48 h. Moreover, dual-targeted LF-NCs showed powerful in vivo antitumor efficacy. It was revealed as significant downregulation of the mRNA expression levels of nuclear factor-kappa B and tumor necrosis factor α as well as suppression of Ki-67 protein expression level in diethylnitrosamine (DEN)-induced HCC mice ($P < 0.05$). Furthermore, dual-targeted LF-NCs attenuated the liver toxicity induced by DEN in animal models. Overall, this study proposes dual-targeted LF-NCs for combined delivery of SFB and QRC as a potential therapeutic HCC strategy.

KEYWORDS: synergistic drug combination, dual active targeting, lactoferrin nanocapsules, lactobionic acid, glycyrrhetic acid, hepatocellular carcinoma



1. INTRODUCTION

Hepatocellular carcinoma (HCC) is the most prevalent hepatic cancer and the third common cause of cancer-induced death globally per year.¹ The most of patients are diagnosed in intermediate or advanced stage of liver cancer where the only option for treatment is systemic chemotherapy. However, conventional chemotherapy has poor response rate with severe

side effects. Sorafenib (SFB, Nexavar), a first-line therapy in advanced HCC treatment, is a multityrosine kinase inhibitor that suppresses cancer cell angiogenesis and proliferation and

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blocks the RAF/MEK/ERK pathway.² Unfortunately, clinical trials showed that SFB prolonged the median survival of HCC patients only ~3 months. Nevertheless, several limitations are associated with its use, such as severe side effects, including diarrhea and hand–foot skin reaction, multidrug resistance, nonselective accumulation in normal cells, fast drug elimination from bloodstream, as well as poor bioavailability due to its low water solubility (60 $\mu\text{g/mL}$).³ Therefore, many attempts were exerted to improve the anticancer efficacy of SFB such as combination with other chemotherapeutics and/or development of nanoparticle (NP)-based targeted drug delivery. Combination therapies could be a potential solution to overcome the tumor drug resistance or synergize the efficacy of conventional chemotherapy through utilizing several mechanisms of action of multiple therapeutics.^{4,5} Furthermore, several targeted drug-delivery systems have been fabricated to eliminate side effects of SFB and enhance its aqueous solubility and bioavailability such as nanosuspension³ and polymeric nanoparticles.⁶

On the other hand, quercetin (QRC), a natural active flavonoid found in vegetables and fruits, has been reported to have anti-inflammatory, antioxidative, and antitumor effects. QRC suppresses the proliferative activity in various types of cancer cells such as pancreatic⁷ and liver cancer cells.⁸ QRC-induced apoptosis of hepatoma cells through activation of caspases and suppression of survivin and B-cell lymphoma 2 has been reported. Moreover, QRC shows antiangiogenic action through downregulation of vascular endothelial growth factor (VEGF) necessary for neo-vessel formation.^{9,10} A previous study has shown that SFB and QRC combination therapy was more effective against HCC compared to single drug treatment. QRC was found to suppress epidermal growth factor, which is mainly responsible for the resistance that some hepatoma cells develop to SFB. Thus, QRC may sensitize HCC cells to SFB through this mechanism and hence this drug combination was hypothesized to provide a potential HCC treatment.¹¹ However, both SFB and QRC have low aqueous solubility resulting in poor bioavailability thereby reducing their potential therapeutic efficacy.

Polymeric nanocapsules (NCs) provide great merits such as high incorporation efficiency of lipophilic drugs within their oily core and sustained drug release. However, synthetic hydrophobic polymers, e.g., poly(lactic-co-glycolic) (PLGA) and polycaprolactone (PCL), commonly used as the shell fabrication material of the NCs were reported to induce some toxicity and/or allergic reactions. Therefore, in our laboratory, we have explored the potential of biocompatible hydrophilic proteins as alternative shell for drug-loaded oily-core NCs. Multireservoir protamine nanocapsules (PRM-NCs) have been fabricated for combined delivery of celecoxib and letrozole to breast cancer cells.¹² PRM-NCs exhibited superior antitumor efficacy against breast cancer MCF-7 cells in vitro and in cancer-bearing mice. Since the hydrophilic cationic protein lactoferrin (LF) has been well reported as a tumor targeting ligand, it was successfully used to modify the surface of several nanocarriers for enhancing their cellular internalization via LF receptor (LFR) endocytosis overexpressed in various tumors such as breast, liver, and lung cancer cells.^{13–15} We have developed layer-by-layer theranostic NCs by electrostatic deposition of the anionic chondroitin sulfate and cationic lactoferrin–quantum dot conjugate onto the oily core coencapsulating celecoxib and honokiol.¹⁶ These LF-decorated theranostic NCs enabled their fluorescence imaging-based tracing in vitro and in vivo and showed enhanced

cytotoxicity against MCF-7 cells and Ehrlich ascites tumor model.

To further improve liver tumor targeting efficiency, hepatic-specific ligands were introduced onto the surface of nanocarriers for targeting specific receptors that are unique for hepatoma cells. Asialoglycoprotein receptors (ASGPRs) are located on human hepatoma cells at a high density that can recognize and bind to galactose moieties of desialylated glycoprotein. Therefore, galactose-bearing lactobionic acid (LA) has been used as a ligand to develop a hepatocyte-specific targeting carrier.^{17,18} Modification of gold nanoparticles-loaded dendrimers with LA has improved their accumulation into hepatic cancer cells thus enabling computed tomography imaging of liver tumors.¹⁹ On the other hand, glycyrrhetic acid (GA), a pentacyclic triterpenoid extracted from liquorice, has been found to specifically bind to GA receptors identified as protein kinase C α more highly abundant on membranes of HCC cells than other normal liver cells. Therefore, GA has been exploited for targeted drug delivery to hepatic cancer cells.^{20,21} Coupling of GA to doxorubicin (DOX)-loaded albumin nanoparticles increased its accumulation in liver tumors compared to nontargeted nanoparticles after injection into hepatoma-bearing mice.²²

In the current study, we propose dual tumor-targeted LF shell-oily core NCs for co-delivery of SFB and QRC for HCC treatment. First, since SFB is insoluble in both aqueous and oily media, SFB was first preformulated in the form of SFB–phospholipid complex (SPC) to increase its solubility and encapsulation into the oil reservoir of NCs. Both SPC and QRC were then entrapped in the oily core of NCs thus providing controlled drug release. Second, for active targeting, a layer of the cationic protein LF was electrostatically deposited onto negatively charged nanoemulsion (NE) for improving their cellular uptake via LFR-mediated endocytosis. Finally, for dual tumor-targeting action, lactobionic acid (LA) and glycyrrhetic acid (GA) were individually conjugated to LF, to enhance cellular internalization of the NCs via additional binding to ASGP-R or GA-R, respectively. Thus, in this study, we have developed two modes of tumor targeting: first, we used only the cationic LF layer (without decoration) to electrostatically coat the anionic oily core. In this case, only single tumor targeting via targeting LF receptors on the surface of hepatoma cells was achieved. Thus, LF layer acted both as a nanocapsule shell for stabilization and as a tumor-targeting ligand. In the second dual-targeting mode, LF was decorated with LA or GA via carbodiimide reaction, which enhanced the tumor-targeting efficiency through binding to LF as well as ASGP or GA receptors on liver cancer cells, respectively. The developed nanocarriers were explored in vitro and in vivo to prove their superior anticancer efficacy compared to the free drug combination therapy.

2. EXPERIMENTAL SECTION

2.1. Materials. All of the chemicals used in this study are detailed in the [Supporting Information](#).

2.2. Preparation of Sorafenib–Phospholipid Complex (SPC). SFB–phospholipid complex (SPC) was prepared according to previously published literature with slight modification.²³ Briefly, SFB and Lipoid S100 were co-dissolved at 1:1 molar ratio in 10 mL of absolute ethanol in a 50 mL conical flask. The mixture was refluxed at 60 °C for 1 h followed by solvent removal under reduced pressure in R-300 model rotary evaporator (Buchi, Switzerland).

2.3. Preparation of LA/LF and GA/LF Conjugates. LA/LF and GA/LF chemical conjugates were prepared via carbodiimide coupling reaction.¹⁷ LA (10 mg) was dissolved in water, followed by addition of

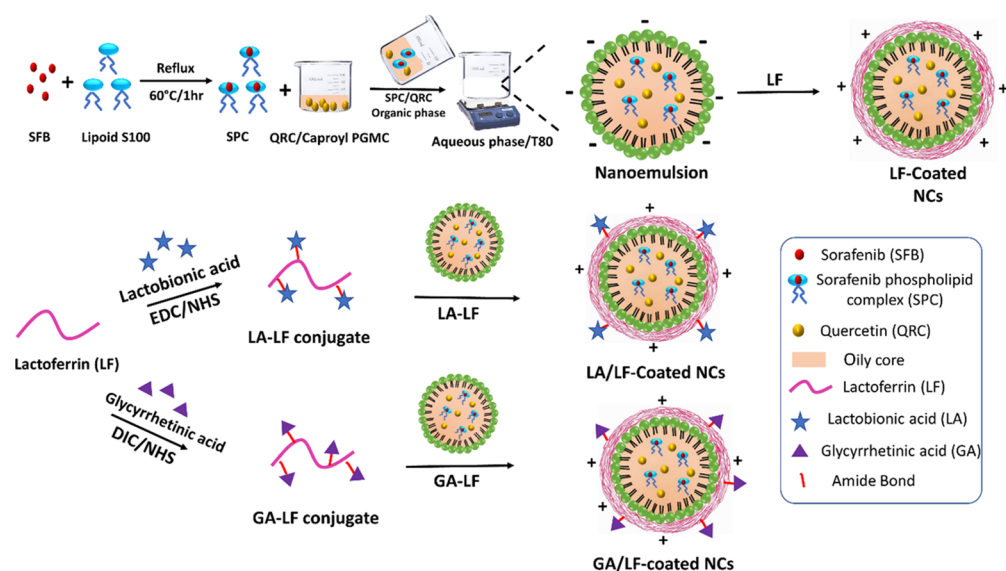


Figure 1. Schematic diagram illustrating the preparation steps of LF-coated NCs.

N-(3-dimethylaminopropyl)-*N*-ethyl carbodiimide hydrochloride (EDC-HCl) and *N*-hydroxysuccinimide (NHS) (LA/EDC/NHS molar ratio, 1:2.4:2.4) and stirred for 1 h for activation. Subsequently, 200 mg of LF was added to the activated aqueous LA solution under magnetic stirring for 24 h at room temperature. The mixture was then dialyzed (12–14 kDa molecular weight cut-off (MWCO) VISKING dialysis tubing, SERVA, Germany) against purified water for 48 h; then, the product was lyophilized.

Similarly, GA/LF conjugate was prepared via carbodiimide coupling reaction while using *N,N'*-diisopropylcarbodiimide (DIC)/NHS coupling reagents.²⁴ GA (12 mg, 0.025 mmol) was dissolved in 2 mL of dimethyl sulfoxide (DMSO), followed by addition of DIC and NHS (GA/DIC/NHS molar ratio, 1:2.4:2.4) and stirred for 1 h. Subsequently, 200 mg of LF was then added to the activated GA solution under magnetic stirring for 24 h at room temperature. The mixture was then dialyzed (12–14 kDa MWCO VISKING dialysis tubing, SERVA, Germany) against pure DMSO for 24 h, while gradually increasing water ratio until dialysis against purified water for 48 h, and then the product was lyophilized.

2.4. Matrix-Assisted Laser Desorption/Ionization Time-of-Flight Mass Spectrometry (MALDI-TOF-MS) Analysis. Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS) analysis was performed to monitor the change in molecular weight of LF before and after LA or GA conjugation. The methodology of MALDI-TOF-MS measurements is described in detail in the [Supporting Information](#).²⁵ Furthermore, the content of glycyrrhetic acid in GA/LF conjugate was determined by indirect method in the dialysate via a reversed-phase high-performance liquid chromatography method at 250 nm²⁴ (details in the [Supporting Information](#)).

2.5. Preparation of Dual Drug-Loaded LF, LA/LF, or GA/LF NCs. LF shell-oily core NCs were fabricated via a two-step spontaneous emulsification/polymer coating technique.²⁶ This technique is mainly based on electrostatic deposition of the cationic LF shell onto the negatively charged nanoemulsion (NE) core. In detail, 5 mg of QRC was dissolved in 0.8 mL of Capryol propylene glycol monocaprylate (PGMC). Ethanolic solution containing SFB–phospholipid complex (SPC, 10 mL), equivalent to 5 mg of SFB, and 50 mg of Lipoid S75 were mixed with the oily phase. The previous organic phase was added to 50 mL of aqueous phase containing Tween 80 (0.2% w/v) under moderate magnetic stirring, resulting in immediate milky NE that formed by virtue of the diffusion of ethanol toward the aqueous phase. The solvent in NE was evaporated under reduced pressure at 40 °C and 75 rpm in a rotary evaporator to reach a volume of 10 mL. Finally, to obtain LF shell-oily core dual drug-loaded NCs, different volumes of 2.5% w/v

aqueous solution of LF, LA/LF, or GA/LF were added to 5 mL of the previously prepared NE under gentle magnetic stirring for 30 min.

2.6. Physicochemical Characterization of LF NCs. Methodologies for evaluating particle size (PS), ζ potential, encapsulation efficiency,^{27,28} in vitro drug release, physical stability, morphological analysis,²⁹ lyophilization redispersibility, solid-state characterization,³⁰ hemolytic compatibility, and serum stability were carried out as previously described³¹ and are detailed in the [Supporting Information](#).

2.7. In Vitro Cytotoxicity and Cellular Uptake Study. The cytotoxicities of the free drugs and dual drug-loaded LF, GA/LF, and LA/LF-NCs on HepG2 liver cancer cells were evaluated via 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) assay³² described and detailed in the [Supporting Information](#). Cellular uptake of the prepared rhodamine B isothiocyanate (RBITC)-coupled LF, GA/LF, and LA/LF-NCs and free dye into HepG2 liver cancer cells was assessed using confocal laser scanning microscopy,^{17,33} as described and detailed in the [Supporting Information](#).

2.8. In Vivo Antitumor Efficacy. The antitumor activity of the prepared dual drug-loaded LF-NCs versus free drugs was assessed in HCC-bearing mice. Briefly, HCC was induced chemically in mice via intraperitoneal injection of diethylnitrosamine (DEN) once weekly for 6 weeks at a dose of 75 mg/kg for 3 weeks, followed by 100 mg/kg for the next 3 weeks.³⁴ The DEN-induced HCC mice were randomly divided into seven groups ($n = 7$) and intravenously (i.v.) injected with the following treatments: free QRC, free SFB, free combined QRC/SFB solution, and combined dual drug-loaded LF-NCs-, LA/LF-NCs-, and GA/LF-NCs-treated groups at a dose of 10 mg/kg QRC and 10 mg/kg SFB, three times weekly for 3 weeks. An additional group of mice were untreated as positive control as well as another group of healthy normal mice that was injected with saline as negative control. The body weights of the mice were measured throughout the treatment period. The mice were sacrificed after treatment for 21 days.

Moreover, excised mice livers were weighed and then cut into sections for histopathological and immunohistochemical analysis and for the assessment of tumor growth biomarkers, including semi-quantitative real-time polymerase chain reaction (RT-PCR) analysis and enzyme-linked immunosorbent assay (ELISA) of liver tissues.¹⁷ Moreover, blood was collected at sacrifice and serum hepatotoxicity markers such as aspartate aminotransferase (AST) and alanine aminotransferase (ALT) were estimated.³² The experimental methods are detailed in the [Supporting Information](#).

2.9. Statistical Analysis. Data analysis is detailed in the [Supporting Information](#).

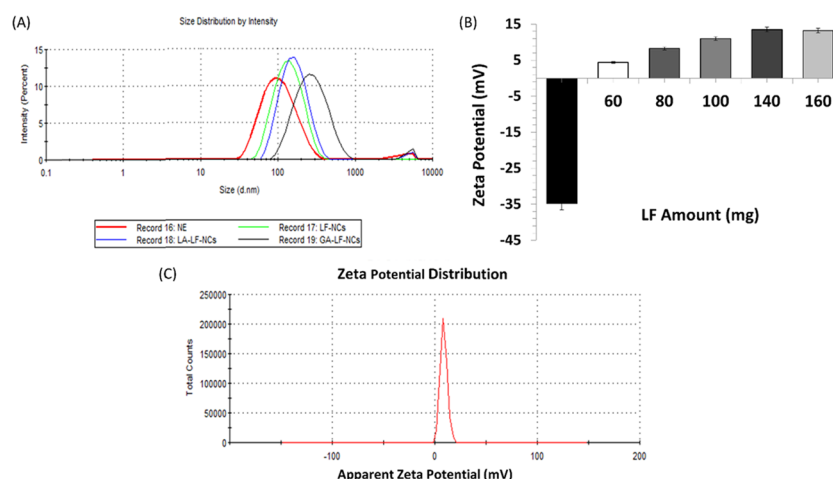


Figure 2. (A) Size distribution diagram of NE, LF-NCs, LA/LF-NCs, and GA/LF-NCs; (B) change in ζ potential upon deposition of LF layer on NE; and (C) ζ potential of LA/LF-NCs.

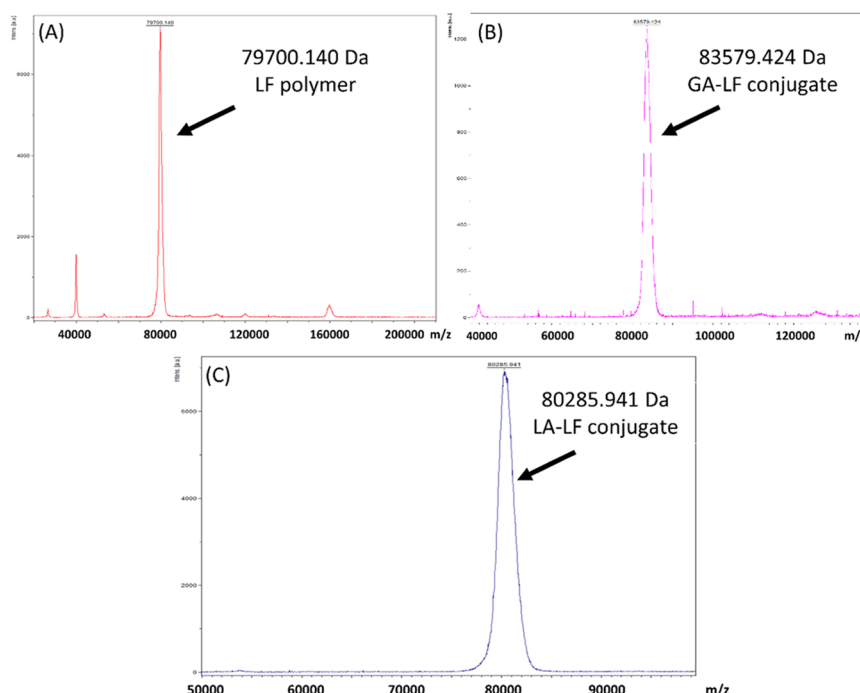


Figure 3. (A) MALDI-TOF-MS analysis of native/control LF, (B) GA-LF conjugate, and (C) LA-LF conjugate.

3. RESULTS AND DISCUSSION

3.1. Development of QRC/SPC-Loaded LF-NCs. We have developed lactoferrin (LF) shell-coated oily core NCs for combined delivery of the two hydrophobic anticancer drugs, sorafenib (SFB) and quercetin (QRC), to liver cancer cells. LF is a naturally occurring cationic glycoprotein present mainly in mammalian excretions, with isoelectric point of ~ 8.7 . Several studies have utilized LF as a targeting ligand for improving cellular uptake of nanoparticles (NPs) into cancer cells via LF-mediated endocytosis. Therefore, besides its biodegradability and biocompatibility, LF is advantageous as a polymeric shell of the NCs compared to the commonly used hydrophobic polymers, e.g., PLGA or PCL. First, the selective tumor targeting properties of LF enhance internalization of the NCs into cancer cells via both cationic charge-facilitated uptake and its preferential binding to multiple receptors overexpressed on hepatoma cells prior to internalization, including the liver

lipoprotein receptor-related protein and asialoglycoprotein (ASGP) receptors.^{32,35} Second, the hydrophilic nature of LF prevents opsonization and enables prolonged circulation of the NCs and hence better tumor accumulation compared to the hydrophobic synthetic polymers. Third, the available functional groups of LF protein structure provides good opportunity for surface functionalization with tumor targeting ligands via simple coupling reactions.³⁶ We have previously prepared LF-coated gliadin nanospheres co-loaded with celecoxib and diosmin for HCC therapy. The LF-coated nanospheres have shown higher cellular uptake and enhanced antitumor effect against liver cancer cells and HCC-bearing mice compared to noncoated gliadin NPs.³²

In the current study, LF shell-coated NCs for co-delivery of SFB and QRC were developed through two steps. First, SFB-phospholipid complex (SPC) was prepared using a simple solvent evaporation method to improve the incorporation of

Table 1. Composition and Physicochemical Characteristics of Dual Drug-Loaded NE and LF Shell-Oily Core NCs

formula	particle size (nm)	PDI	ζ -potential (mV)	EE % (w/w)	
				SFB	QRC
blank NE	92.64 $\pm$ 1.5	0.211	−34.0 $\pm$ 0.21		
NE	112.3 $\pm$ 2.8	0.241	−30.9 $\pm$ 0.36	80.8 $\pm$ 1.50	87.0 $\pm$ 0.45
LF-NCs	146.1 $\pm$ 2.2	0.149	+13.5 $\pm$ 0.25	82.5 $\pm$ 0.91	83.0 $\pm$ 0.65
LA/LF-NCs	169.0 $\pm$ 1.5	0.218	+8.81 $\pm$ 0.51	85.6 $\pm$ 0.84	86.5 $\pm$ 0.56
GA/LF-NCs	230.2 $\pm$ 1.7	0.220	+14.4 $\pm$ 0.22	88.0 $\pm$ 0.32	88.0 $\pm$ 0.22

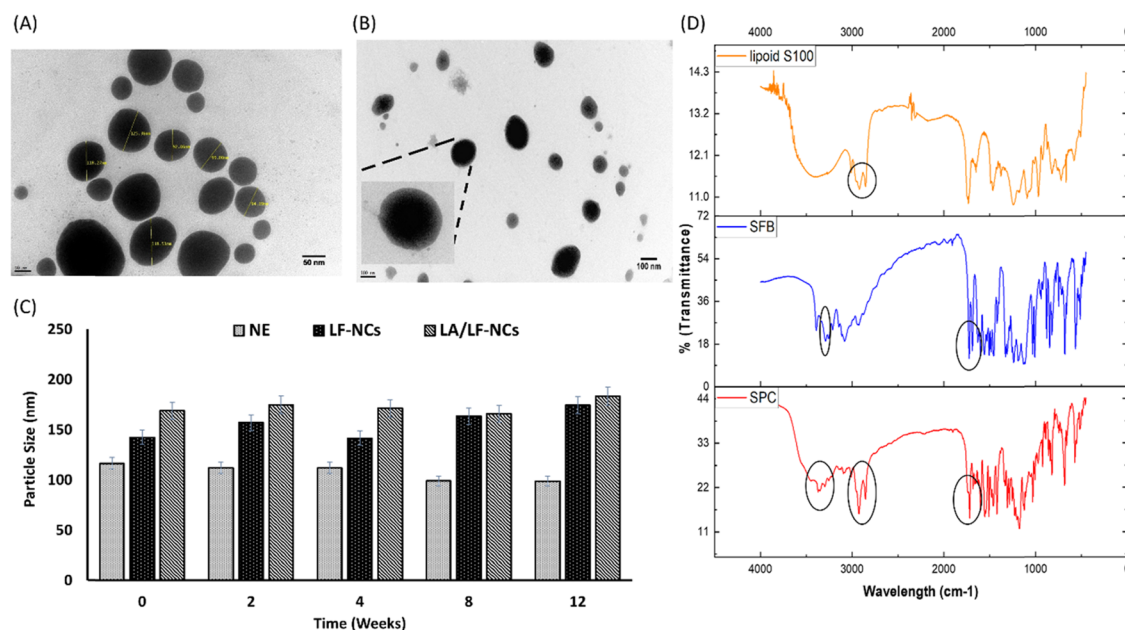


Figure 4. (A) TEM image showing morphology of NE, (B) TEM image showing morphology LF-coated NCs, (C) physical stability of NE, LF-NCs, and LA/LF-NCs showing the change in particle size with time, (D) Fourier transform infrared (FTIR) spectra of Lipoid S100, SFB, and SPC complex.

SFB within the oil phase. SPC/QRC dual drug-loaded anionic oily-core nanoemulsion (NE) was then prepared via o/w spontaneous emulsification/solvent displacement technique upon adding the organic phase to aqueous phase containing Tween 80 as emulsifying agent. The preparation steps of LF shell-oily core NCs are illustrated in the schematic diagram shown in Figure 1. In our preliminary studies, different types of oils were investigated for preparation of the NE. Capryol PGMC was finally selected based on its high drug solubilizing and emulsion stabilizing capacity. For enhancing the NE stability, two surfactants, hydrophilic Tween 80 and lipophilic Lipoid S75, were involved. The resultant QRC/SPC-loaded NE exhibited PS of 112.3 nm (Figure 2A), with a highly negative ζ -potential of −30.9 mV, which was attributed to the phosphatidic acid and free fatty acids in phospholipids. This was followed by coating of the negatively charged NE with the cationic LF solution through electrostatic interaction to develop NCs.¹² Upon coating of NE using different concentrations of LF solution, the ζ -potential was increased gradually and reversed reaching +13.5 mV at 1.4% w/v with the PS increased from 112.3 to 146.1 nm, thus confirming the deposition of LF layer (Figure 2A–C). These results were in agreement with the size increase and charge reversal upon coating of the anionic chondroitin sulfate-based oily core NCs with outer LF shell.¹⁶

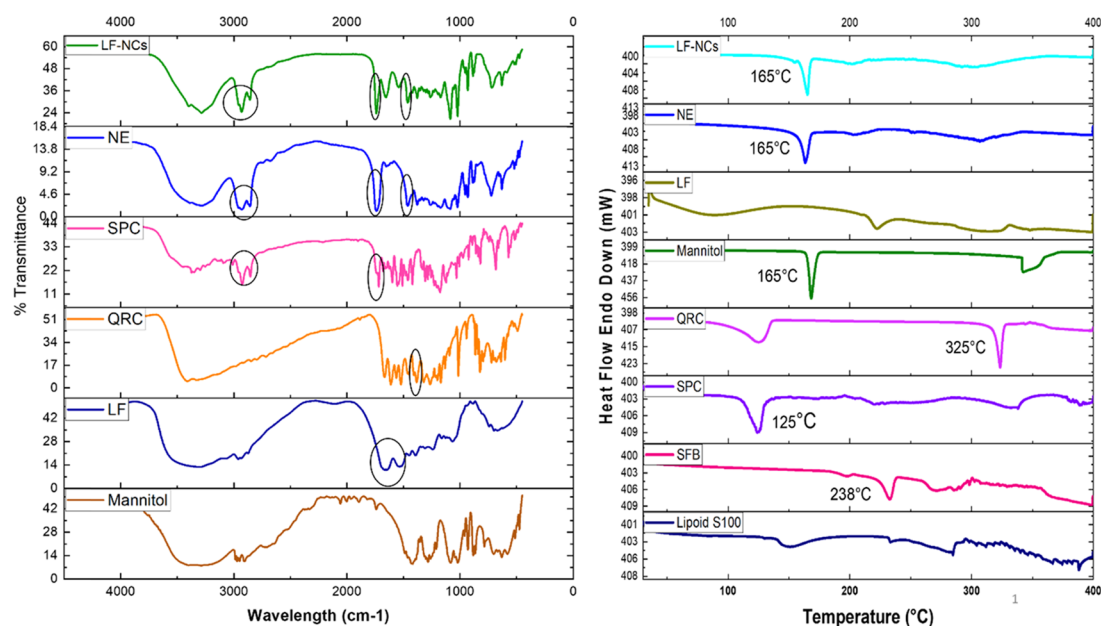
3.2. Design of LA/LF and GA/LF Conjugates and Dual-Targeted LF NCs. For dual tumor targeting, lactobionic acid (LA) or glycyrrhetic acid (GA) was conjugated to LF surface via carbodiimide coupling reaction between the carboxylic

group of LA or GA and free amino groups of LF using NHS-ester-mediated conjugation. LA and GA decoration has been utilized for targeting ASGP and GA receptors, respectively, besides the intrinsic targeting action of LF for liver cancer cells. The changes in molecular weight of LF were determined via MALDI-TOF-MS, as shown in Figure 3A–C. Conjugation of LA (358.296 Da) or GA (470.68 Da) with LF will result in an increase in the molecular weight of LA–LF or GA–LF conjugates compared to LF alone (82 823.375 Da), accompanied by the loss of H₂O molecule. This accounts for conjugation of approximately two molecules of LA in the case of LA–LF conjugate and approximately eight molecules of GA in the case of GA–LF conjugate.

For developing dual-targeted NCs, the synthesized LA–LF and GA–LF conjugates were used for coating the QRC/SPB-loaded NE. Thus, LA–LF and GA–LF NCs with sizes of 169.0 and 230.0 nm, respectively, were elaborated (Table 1 and Figure 2A). This significant increase in particle size can be correlated to the surface modification of LF with targeting moieties. Although the positive charge of our nanocarriers can result in side effects and rapid clearance from body circulation, the hydrophilic glycol part of LF (being a glycoprotein) has been reported to increase the stability of LF nanocarriers by acting like poly(ethylene glycol) (PEG) stealth layer, which minimizes the opsonization and hence attenuates the interaction with body tissues, reduces the rapid clearance of LF-NCs, and prolongs their systemic circulation.^{32,37–39} On the other hand, upon reaching the tumor site, the positive charge of nanocarriers can enhance their

Table 2. Effect of Freeze-Drying on the Physicochemical Characteristics of QRC/SFB-Loaded LF Shell-Oily Core NCs Lyophilized Using 2.5 and 5% w/w Mannitol as Cryoprotectant

	particle size (nm)			PDI			RI ^a (Sf/Si)		yield (% w/w)	
	after			after						
	before	2.5%	5%	before	2.5%	5%	2.5%	5%	2.5%	5%
LF-NCs	149.0	169.3	221.9	0.241	0.31	0.387	1.13	1.48	87.5	91.8
LA/LF-NCs	169.0	195.7	235.9	0.218	0.353	0.452	1.15	1.39	73.0	79.5

^aRI: redispersibility index.**Figure 5.** (A) Fourier transform infrared (FTIR) spectra of mannitol, LF, QRC, SPC complex, NE, and LF-NCs. (B) DSC thermograms of Lipoid S100, SFB, SPC complex, QRC, mannitol, LF, NE, and LF-NCs.

internalization into cancer cells via binding to the negatively charged proteoglycans on the surface of cancer cells in addition to the receptor-mediated endocytosis (RME).^{24,37} High encapsulation efficiency was achieved (up to 85.6 and 86.5% for SFB and QRC, respectively) in all NCs due to their high lipophilicity (Table 1). Similar high encapsulation efficiencies for both drugs were previously observed upon their loading into lipid-coated nanoparticles.⁴⁰

3.3. Morphological Analysis, Physical Stability, and Redispersibility. Transmission electron microscopy (TEM) examinations for the prepared NE and LF-NCs exhibited spherical shape with size range of 95–118.27 nm for NE and 128–168 nm for NCs, without any aggregations, which confirmed their excellent colloidal stability (Figure 4A,B). Moreover, TEM images of LF-coated NCs exhibited core–shell structure, which provides a great evidence for efficient coating of the oily core NE with a hydrophilic shell of LF.¹² As shown in Figure 4C, it was noted that the nanoformulations retained their physicochemical characteristics exhibiting PS values of 112.8, 175.5, and 183.6 nm for NE, LF-NCs, and LA/LF-NCs after storage in colloidal form for 3 months at 4 °C, displaying no significant increase in PS. This excellent physical stability of these nanocarriers can be attributed to using two types of surfactants, hydrophilic Tween 80 and hydrophobic Lipoid S75, thus providing efficient steric stabilization for NCs. Moreover, the cationic LF shell provides charge-based stabilization in addition to the steric stabilization induced by its glycol part of the glycoprotein chain that can prevent coalescence of NCs.⁴¹

Similar excellent colloidal stability was reported for chondroitin sulfate-based NCs upon coating with an LF corona.¹⁶

Lyophilization is considered a common technique for prolonging the physical stability of NCs upon storage. Since the oily core of NCs is susceptible to collapse upon drying forming sticky nonredispersible powder, the presence of a cryoprotectant is essential for freeze-drying of NCs. In our study, using 2.5% w/v mannitol as a cryoprotectant successfully hindered the aggregation of NCs and maintained their original size after freeze-drying. Moreover, LF- and LA/LF-NCs displayed redispersibility indexes of 1.13 and 1.15, respectively, indicating their high reconstitution properties (Table 2). Mannitol was previously used to protect oily core NCs against aggregation during the freeze-drying process.³⁰

3.4. Solid-State Characterization. FTIR spectroscopy for SPC was performed to understand the interaction between SFB and phosphatidylcholine molecules (Lipoid S100) (Figure 4D). FTIR spectrum of SPC showed characteristic peaks at 2922 and 2852 cm⁻¹ assigned as C–H stretching band of phosphatidylcholine, in addition to the absorption band corresponding to C–F bending band of SFB at 686 cm⁻¹.^{38,39} In addition, the absorption band of SFB at 3391 cm⁻¹ that corresponds to N–H amide group was observed in the SPC spectrum at 3372 cm⁻¹. Moreover, all carbonyl groups of SFB and phospholipid overlapped and shifted to 1718 cm⁻¹ in the SPC spectrum. The observed shift might be due to the hydrogen bonding that occurred between SFB and phosphatidylcholine.³⁹

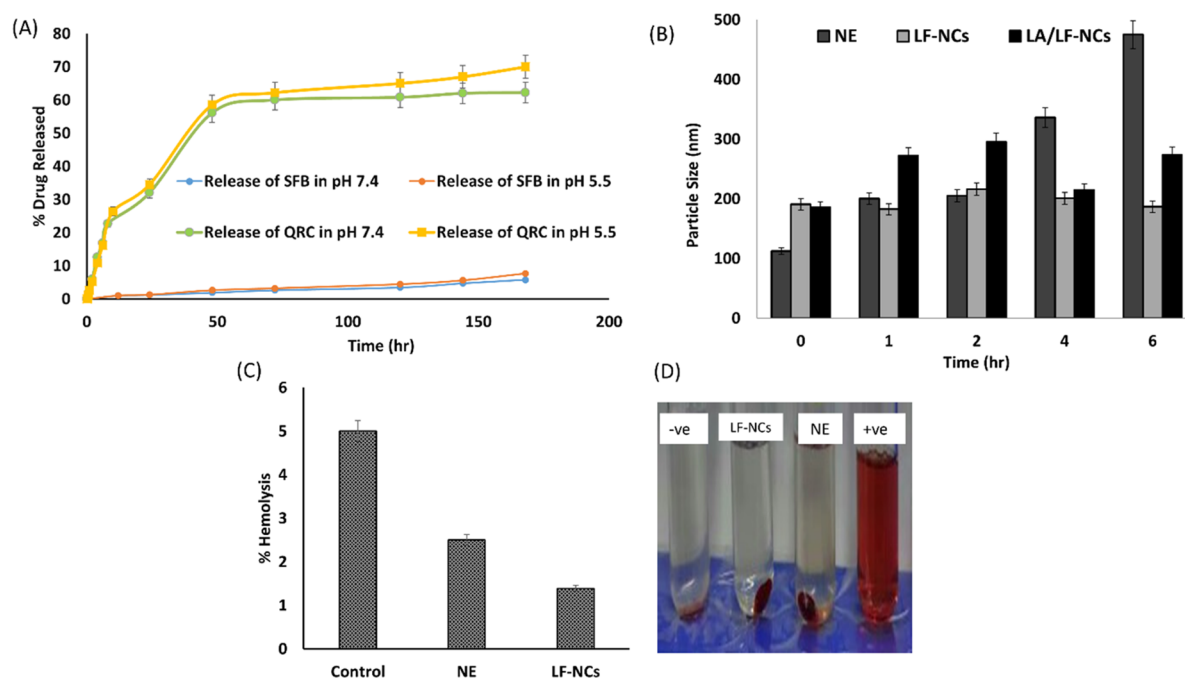


Figure 6. (A) In vitro release of SFB and QRC from the dual drug-loaded NCs using 100 mL of phosphate-buffered solution containing 1% (w/v) Tween 80 as a release medium at 37 °C, 100 rpm; (B) serum stability of NE, LF-NCs, and LA/LF-NCs after incubation in 10% fetal bovine serum (FBS) for 6 h at 37 °C; (C) hemolytic potential of dual drug-loaded NE and NCs showing % hemolysis; and (D) hemocompatibility image after 1 h of incubation with RBC at 37 °C.

The FTIR spectra (Figure 5A) of NE and LF-NCs have revealed absorption bands at 2930, 2860, and 1730 cm^{-1} , which accounted for successful SPC encapsulation. QRC major characteristic absorption bands for C=O stretching at 1667 cm^{-1} and C—O—C stretching at 1316.49 and 1163.08 cm^{-1} were also detected in IR spectra of NE and LF-NCs.³⁷ It has been noted that LF-NCs spectrum displayed the main characteristic stretching and bending vibrations of LF as amide I (1636.49 cm^{-1}) and amide II (1541.29 cm^{-1}), thus confirming the LF deposition onto the oily core NE.⁴⁰ The differential scanning calorimetry (DSC) thermograms of free QRC, free SFB, SFB–phospholipid complex (SPC), lyophilized NE, and LF-NCs are illustrated in Figure 5B. The thermogram of SFB–phospholipid complex (SPC) showed disappearance of the endothermal peaks of SFB (at 238 °C) and phospholipid (at 152 °C) accompanied with the appearance of a new endothermal peak at 125 °C. This result confirmed the formation of phospholipid complex associated with some sort of interactions between SFB and phospholipids, such as hydrogen bonds and van der Waals force.⁴² The DSC thermograms of QRC and SPC revealed two sharp endothermic peaks corresponding to their melting points at 325 and 125 °C.^{5,43} However, these peaks have disappeared in the thermograms of both NE and LF-NCs, which confirms the presence of both drugs in amorphous phase in the carrier matrix. The thermograms of the two lyophilized nanoformulations (NE and LF-NCs) revealed a sharp endothermic peak at 165 °C corresponding to melting point of mannitol used as a cryoprotectant.

3.5. In Vitro Release Study. The in vitro release profiles of SFB or QRC at pH values 7.4 and 5.5 from LF-coated NCs are depicted in Figure 6A. The results showed that the release of SFB and QRC from LF-coated NCs at pH values 7.4 and 5.5 showed no obvious difference. Nearly 6% of SFB was released from NCs approximately in 7 days, while about 60% of QRC was

released after 24 h. The reason behind this prolongation of release time is related to the oily core that acts as depot for both hydrophobic drugs.⁴⁴ The release rate of SFB from NCs was remarkably lower than that of QRC, which may be explained by its higher log *P* value (3.8 vs 1.82 for SFB and QRC, respectively) besides its additional complexation with phospholipid.^{45,46} Therefore, LF-coated NCs have efficiently achieved a sustained release for SFB, which can increase its safety issue through avoiding its leakage in blood after i.v. administration. Moreover, this sequential release pattern was beneficial so that the released QRC would sensitize HCC cells to SFB action thus enhancing its antitumor efficacy.¹¹

3.6. Serum Stability and Hemocompatibility. All of the prepared NCs exhibited increase in their particle size upon mixing with FBS (Figure 6B), where the sizes of NE, LF-NCs, and LA/LF-NCs were increased to 200.3, 182.4, and 222.0 nm, respectively, after 1 h incubation. This size increase was mainly attributed to the binding of serum proteins to NCs surface developing a protein corona. Further size increase was noted after 2 h due to the continuous accumulation of serum proteins. Then, a drastic decrease in size was observed for LF-NCs and LA/LF-NCs after 6 h, reaching 186.8 and 215.5 nm, respectively, as a result of desorption of serum proteins. Besides, the effect of osmotic pressure of serum proteins causes leakage of water from the aqueous part of NCs resulting in their size shrinkage. Similar results were previously observed upon using positively charged proteins as gelatin and LF as the fabricating shell of NCs.^{16,30} However, the size of uncoated NCs (NE) has been further increased with time reaching 475.0 nm after 6 h, due to the absence of the hydrophilic protein shell that prevents coalescence of oil droplets resulting in hydrophobic interactions.⁴⁷

On another avenue, the hemolytic activity of dual drug-loaded NE and LF-NCs was low and negligible, 2.5 and 1.39%, respectively, up to a concentration of 1 mg/mL (Figure 6C,D).

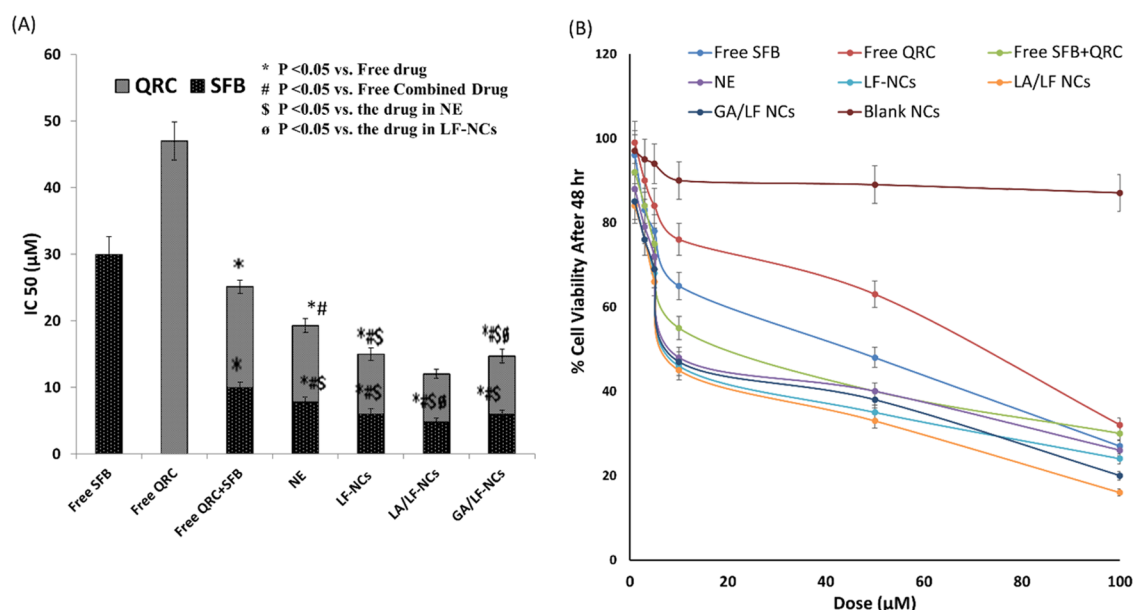


Figure 7. (A) Cytotoxicity analysis of free QRC, free SFB, and free QRC/SFB co-solvent compared to NE, LF-NCs, LA/LF-NCs, and GA/LF-NCs on HepG2 liver cancer cell line at the concentration of 0–100 μM after 48 h. (B) IC₅₀ of free drugs and different nanoformulations.

Table 3. CI, IC₅₀, and DRI Values of Free Drugs Compared to the Prepared QRC/SFB-Loaded LF Shell-Oily Core NCs against HepG2 Liver Cancer Cells at the Concentration of 0–100 μM after 48 h

drug/combo	CI value	total IC ₅₀ of combination	dose SFB	dose QRC	DRI of SFB	DRI of QRC
free SFB			29.96			
free QRC				47.02		
free QRC + SFB	0.952	25.1007	9.97	15.13	3.01	3.11
NE	0.918	19.2709	7.79	11.48	3.84	4.09
LF-NCs	0.897	15.0139	6.00	9.01	4.98	5.21
GA/LF-NCs	0.873	14.6983	5.92	8.77	5.06	5.36
LA/LF-NCs	0.836	11.8290	4.81	7.21	6.22	6.52

Generally, a hemolysis percentage of <5% is considered as nontoxic and safe.⁴⁵ These results indicated that dual drug-loaded NE and LF-NCs had good hemocompatibility, and hence they were suitable for parenteral administration.

3.7. In Vitro Cytotoxicity and Cellular Uptake Study.

The cytotoxicities of free drugs and QRC/SPC co-loaded NCs against HepG2 liver cancer cells were investigated by 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) assay (Figure 7A,B). First, the noncytotoxicity of blank NCs against HepG2 cells after 48 h was confirmed, revealing their biocompatibility and safety. Compared to free combination therapy (QRC + SFB), single-targeted QRC/SPC-LF NCs displayed slightly higher cytotoxicity in both 24 and 48 h incubation. The half-maximal inhibitory concentration (IC₅₀) value of NE was 1.3-fold lower than free drug (QRC + SFB) combination after 48 h. LF-NCs and GA/LF-NCs had no significant difference in their IC₅₀ values, which were 1.7-fold lower than free drug combination after 48 h. However, dual-targeted LA/LF-NCs showed twofold reduction in IC₅₀ compared to free combination therapy after 48 h against HepG2, which was likely due to increased cellular internalization as mentioned above.

In addition, further extensive statistical analysis was performed via CompuSyn software that was described by Chou and Talalay.^{48–50} We estimated the combination index (CI) and dose reduction index (DRI) to evaluate the antitumor efficacy of the different NCs correlated to the free combination

therapy (Table 3). Our results revealed that all prepared nanocarriers have superior cytotoxic activity compared to free combination, especially GA/LF-NCs and LA/LF-NCs, whose CIs were 0.873 and 0.836, respectively, thus confirming the synergistic effect achieved through co-loading of QRC/SFB in targeted NCs. Furthermore, the DRIs of SFB were 5.06 and 6.22 in GA/LF-NCs and LA/LF-NCs, respectively. DRIs of QRC were 5.36 and 6.52 in GA/LF-NCs and LA/LF-NCs, respectively. The enhanced cytotoxicity of LA/LF-NCs was attributed to the targeting action of LA that targets ASGP receptor, besides LF targetability. On the contrary, the conjugation of GA to LF did not show the same cytotoxic effect of LA due to the steric hindrance of the attached hydrophobic GA moieties that may affect receptor-binding ability of GA/LF-NCs. Similarly, Wang et al. reported that norcantharidin-associated galactosylated chitosan NPs demonstrated higher cellular uptake and strong cytotoxicity against HepG2 cells compared to nongalactosylated NPs.⁵¹

To investigate the cellular internalization of NCs, rhodamine B isothiocyanate (RBITC)-labeled LF conjugates were prepared through direct conjugation reaction between isothiocyanate group of RBITC and free amino groups of LF. RBITC-LF conjugates were used to coat NE forming RBITC-labeled LF-NCs, which would be incubated with HepG2 cells then visualized via confocal laser scanning microscope. As shown in Figure 8a,b, the three prepared NCs displayed brighter red fluorescence intensity in cytoplasm of cells compared to free

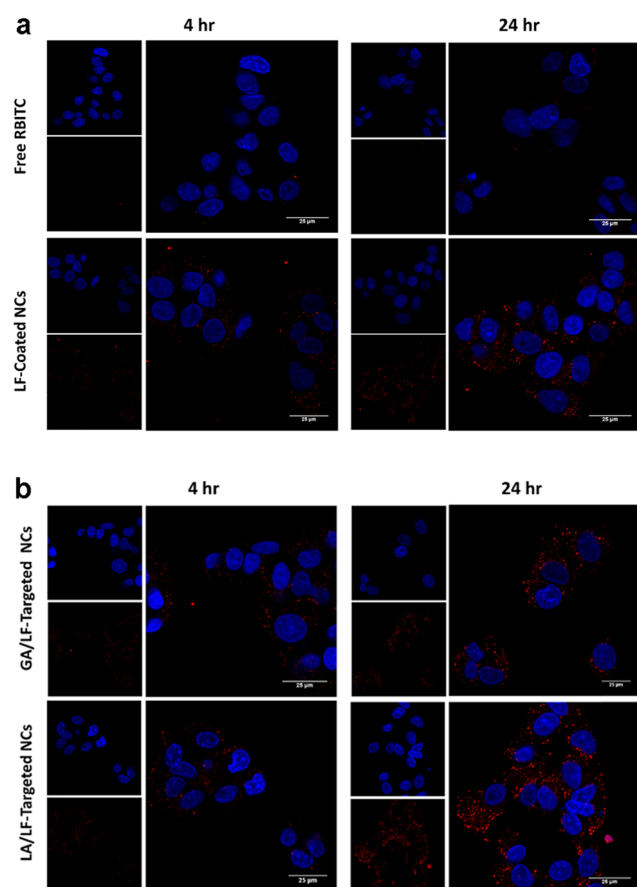


Figure 8. Confocal images showing cellular uptake of (a) free RBITC and LF-NCs and (b) LA/LF-NCs and GA/LF-NCs within HepG2 liver cancer cells after incubation for 4 and 24 h.

RBITC, confirming the enhanced cellular uptake of NCs. The intracellular fluorescence intensity increased with time, indicating that the cellular internalization of the three prepared NCs was time-dependent. However, LF-NCs and GA/LF-NCs demonstrated comparable fluorescence intensity, whereas LA/LF-NCs revealed the strongest fluorescent signals after 4 and 24 h incubation. The enhanced cellular uptake of our positively charged NCs could be due to the electrostatic interaction with negatively charged cell membranes that augment their penetration via adsorptive-mediated transcytosis. In addition, the ligands LF, LA, and GA can enhance cellular internalization of NCs via receptor-mediated endocytosis (RME) through binding to LF, ASGP, and GA receptors overexpressed on the surface of HepG2 cells.^{24,37} The HepG2 hepatoma cells express 76 000 ASGPRs/cell with a high density on the membrane.⁵² These receptors can recognize and bind galactose residues of LA with high specificity and efficiency. Moreover, HepG2 cells have overexpressed GA receptors on their surface compared to other cell lines.^{22,53,54} On the contrary, the conjugation of GA to LF did not show the same enhanced uptake behavior of LA, which may be due to the steric hindrance of the attached hydrophobic GA moieties that can affect the receptor-binding ability of GA/LF-NCs. Moreover, the bigger size of GA/LF-NCs (230.2 ± 1.7 nm) compared to LA/LF-NCs (169.0 ± 1.5 nm) may also contribute to its lower internalization capacity.

3.8. In Vivo Antitumor Efficacy. In previous studies, LF has been successfully exploited for targeted delivery of anticancer drugs to liver cancer cells. LF nanoparticles loaded

with DOX were used for treatment of diethylnitrosamine-induced HCC in rats.^{55,56} Administration of LF nanocarriers into the rats decreased the number of liver tumor nodules by >93% with higher drug accumulation in liver compared to the free one. In addition to be used as the main carrier, LF was also used as a targeting ligand decorating the surface of liposomal or polymeric nanocarriers for targeted drug delivery to liver cancer cells. When LF was coupled to the surface of PEGylated liposomes encapsulating a fluorescent dye, higher accumulation in liver cancer cells was revealed both in vitro and in vivo.¹³ Similarly, in our laboratory, we found that coating of gliadin nanospheres with LF resulted in enhanced delivery of celecoxib and diosmin to liver cancer cells via binding to ASGP receptors, which was reflected as reduced expression of tumor angiogenic and inflammatory markers as well as induction of caspase-mediated apoptosis.³² Therefore, in the current study, we evaluated the tumor-targeting effect of single and dual-targeted LF shell-coated NCs in HCC-bearing mice.

3.8.1. Assessment of Tumor Growth Biomarkers Expression Levels. **3.8.1.1. Evaluation of Nuclear Factor-Kappa B (NF- κ B) Expression Level.** Nuclear factor-kappa B (NF- κ B) has a vital role in the development and progression of HCC; thus, it is considered as an attractive target for treating HCC. A previous study reported that inhibition of NF- κ B expression highly sensitized hepatoma cells to SFB-induced apoptosis.⁵⁷ With regard to NF- κ B expression, Granado-Serrano et al. reported that QRC has suppressed the nuclear translocation of NF- κ B in liver cancer cells.⁸ Accordingly, the combination of QRC with SFB may enhance its antitumor efficacy. In our study, free QRC + SFB combination showed no significant difference in lowering NF- κ B mRNA expression, which may be due to their high lipophilicity and poor penetration through tumor tissue (Figure 9A,B), while QRC + SFB coencapsulated in GA/LF-NCs has revealed significant reduction in NF- κ B expression level compared to positive control, which can be due to their enhanced uptake through LF-mediated endocytosis ($P < 0.05$). Furthermore, LA/LF-NCs displayed the lowest NF- κ B mRNA

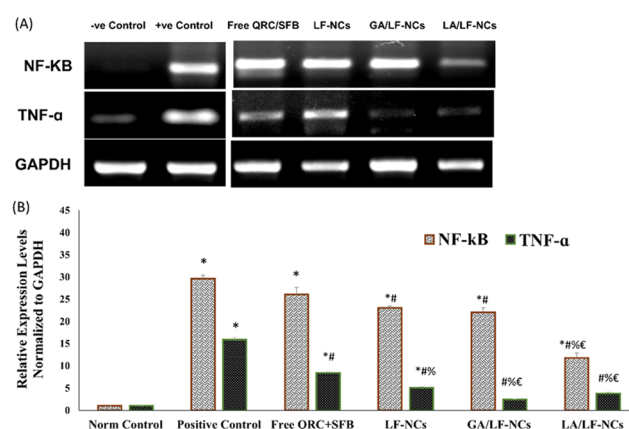


Figure 9. (a) Expression profile for mRNA levels of NF- β B and tumor necrosis factor α (TNF- α) in groups treated with free combined, LF-NC, LA/LF-NC, and GA/LF-NC individual tissues measured by (RT-PCR) and normalized to glyceraldehyde-3-phosphate dehydrogenase (GAPDH) ($n = 4$). (b) Quantitative expression of NF- κ B and TNF- α levels quantified by RT-PCR for the studied groups, normalized according to GAPDH ($n = 4$). (* $P < 0.05$ vs negative control, # $P < 0.05$ vs positive control, % $P < 0.05$ vs free QRC/SFB, $^{\epsilon}$ $P < 0.05$ vs LF-NCs, $^{\$}$ $P < 0.05$ vs GA/LF-NCs.)

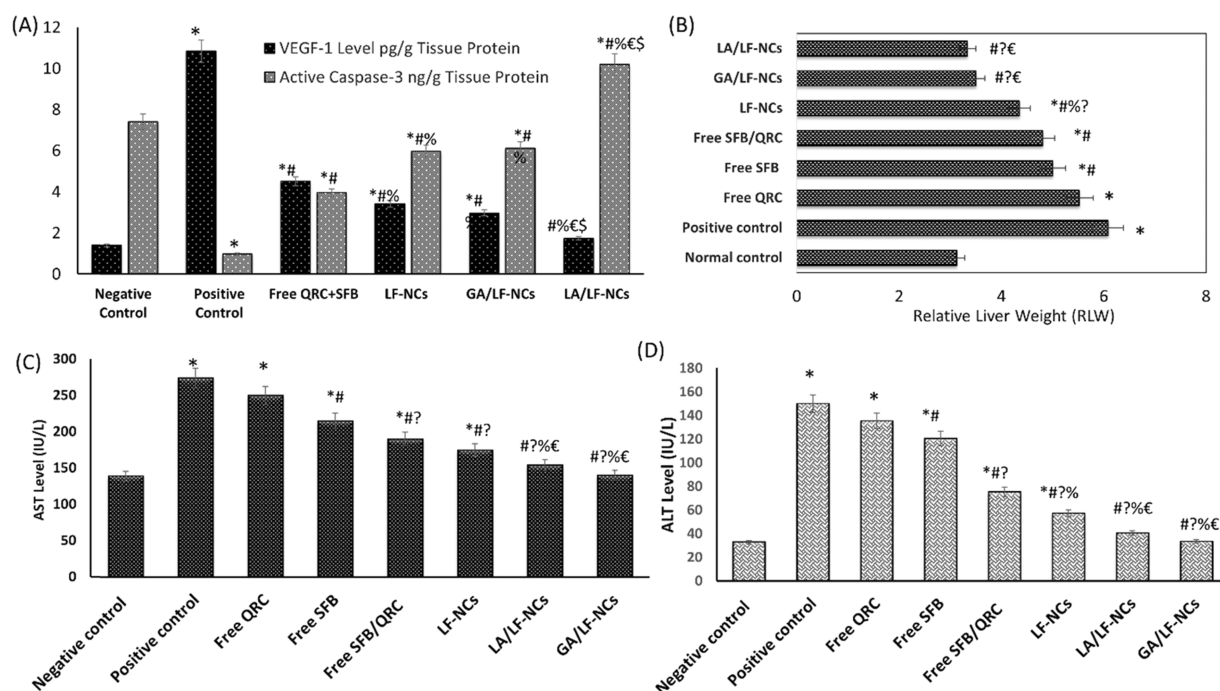


Figure 10. Comparison between the studied groups (free QRC, free SFB, and free QRC/SFB co-solvent compared to NE, LF-NCs, LA/LF-NCs, and GA/LF-NCs)-treated groups in addition to the positive control group according to (A) VEGF-1 level and active caspase-3, (B) Relative liver weight (RLW%), (C) ALT level, and (D) AST level ($n = 4$). (* $P < 0.05$ vs negative control, # $P < 0.05$ vs positive control, ? $P < 0.05$ vs free SFB, % $P < 0.05$ vs free QRC/SFB, € $P < 0.05$ vs LF-NCs, \$ $P < 0.05$ vs GA/LF-NCs.)

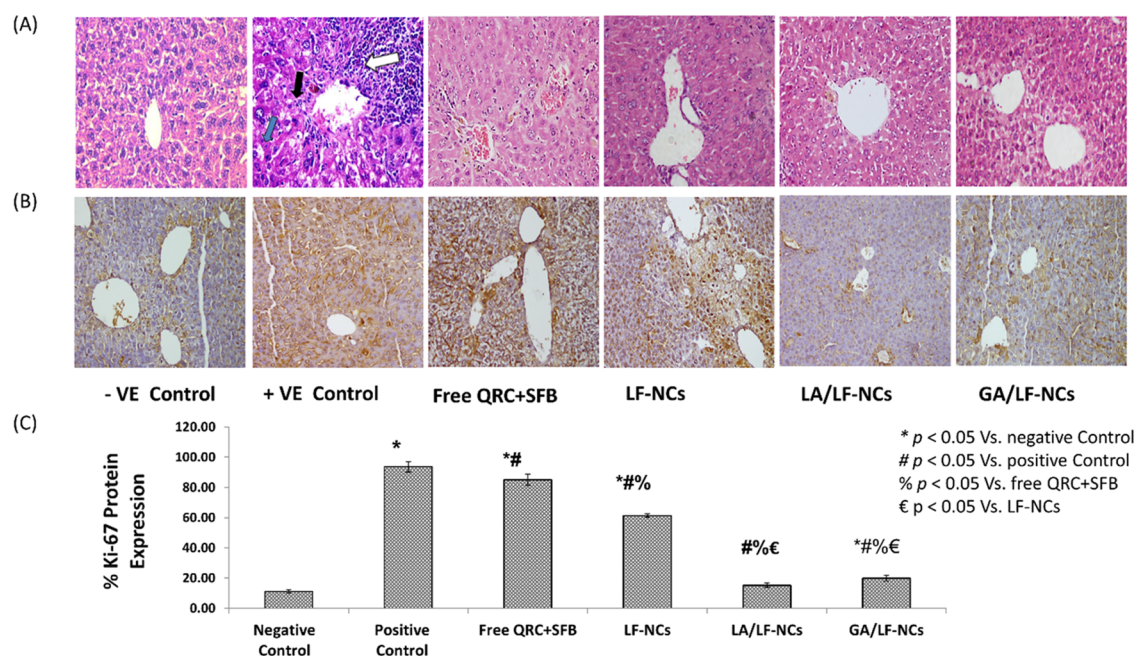


Figure 11. (A) Hematoxylin and eosin staining of liver cancer tissues of negative control, positive control group, free QRC/SFB combination, LF-NCs, LA/LF-NCs, and GA/LF-NCs-treated groups. (B) Immunohistopathological staining of the proliferative marker Ki-67 in liver cancer tissues of positive control group and free QRC/SFB combination, LF-NCs, LA/LF-NCs, and GA/LF-NCs-treated groups and (C) % Ki-67 proliferation marker in positive control group and liver cancer tissues of free QRC/SFB combination, LF-NCs, LA/LF-NCs, and GA/LF-NCs-treated groups ($n = 4$). (* $P < 0.05$ vs negative control, # $P < 0.05$ vs positive control, % $P < 0.05$ vs free QRC/SFB, € $P < 0.05$ vs LF-NCs, \$ $P < 0.05$ vs GA/LF-NCs.)

expression level which was attributed to their improved uptake via both LF and ASGP receptors.

3.8.1.2. Evaluation of Tumor Necrosis Factor α (TNF- α) Expression Level. Tumor necrosis factor (TNF)- α is considered as a crucial inflammatory mediator overexpressed in hepatoma cells more than normal hepatocytes and promoting tumor

growth and invasion.⁵⁸ A previous study has shown that overexpression of TNF- α was accompanied with low response to SFB.⁵⁹ However, Nair et al. reported in his study that QRC suppresses TNF- α gene and protein expression via knocking down NF- κ B gene expression.⁶⁰ Therefore, we have administered SFB in a combination with QRC in an attempt to enhance

the antitumor efficacy of SFB. In the current study, free QRC/SFB combination demonstrated significant suppression in mRNA expression level of TNF- α compared to positive control ($P < 0.05$) (Figure 9A,B). Moreover, GA/LF-NCs and LA/LF-NCs both have displayed the lowest TNF- α expression level compared to free QRC/SFB combination ($P < 0.05$), which correlated to the targeting action of both NCs.

3.8.2. Detection of Antiangiogenic Effect. SFB has an antiangiogenic action through direct effect on vascular endothelial growth factor (VEGF).² Also, previous studies showed that QRC has suppressed neovessel density and reduced VEGF expression in HCC-induced animals.^{9,61} Thus, QRC may synergize the action of SFB against HCC cells.¹¹ In our study, free combined drug-treated group displayed significant suppression to VEGF protein expression level compared to positive control group ($P < 0.05$) (Figure 10A). Moreover, LF-NCs- and GA/LF-NCs-treated groups showed similar reduction in VEGF expression level, which was significantly lower than free combination-treated group ($P < 0.05$). Furthermore, LA/LF-NCs exhibited the lowest VEGF expression protein level, which revealed its superior antiangiogenic effect over the other prepared NCs.

3.8.3. Measurement of Apoptosis Induction. Previous reports have shown that QRC inhibited cell proliferation and induced apoptosis attributed to caspase activation in liver cancer cells.⁶² Also, SFB promoted strong activation of the proapoptotic caspase-3 in hepatoma cells.⁶³ In this study, we found that free combined-treated group exhibited higher caspase-3 expression level compared to positive control ($P < 0.05$) (Figure 10A). However, encapsulating both drugs in LF-NCs or GA/LF-NCs showed significant increase in caspase-3 expression level compared to free combination therapy ($P < 0.05$). Furthermore, LA/LF-NCs displayed significant upregulation to the expression level of caspase-3 compared to all other treated groups ($P < 0.05$).

3.8.4. Assessment of Serum ALT and AST Levels and Relative Liver Weight (RLW%). DEN treatment significantly increased the relative liver weight to 6.18 ± 0.15 versus 3.17 ± 0.2 ($P < 0.05$, Figure 10B) compared to negative control group. Also, DEN-treated group demonstrated remarkable increase in serum ALT (149.9 ± 2.7 IU/L) and AST (274.4 ± 4.5 IU/L) levels compared to normal control group (32.8 ± 1.23 and 139.2 ± 2.48 IU/L, respectively; Figure 10C,D). The increase of ALT and AST activities was a result of hepatocellular damage occurred by using DEN. However, LF-NCs-treated groups showed significant amelioration in ALT level by 1.3-fold compared to free QRC + SFB, while GA/LF-NCs and LA/LF-NCs have significantly reduced RLW and serum levels of ALT and AST compared to free combination therapy ($P < 0.05$), which could be comparable to normal level ($P < 0.05$).

3.8.5. Histopathological Study. The histological examinations of liver sections confirmed the results obtained from tumor biomarkers analysis. Negative control group showed normal architecture with central vein and polygonal hepatocytes with granulated cytoplasm (Figure 11A), while liver sections from DEN-treated group revealed loss of architecture and neoplastic cells with larger hyperchromatic nuclei.^{17,64} Furthermore, this section showed irregular sinusoids and infiltration of inflammatory cells. Interestingly, free combined drugs-treated group started to regain the hepatic architecture to limited extent as the Kupffer cells still infiltrated. Moreover, the liver sections obtained from LF-NCs and GA/LF-NCs displayed apparent improvement in the histological features of DEN-treated groups.

Furthermore, LA/LF-NCs-treated groups exhibited remarkable amelioration to the cellular architecture, which was almost comparable to the normal control liver section.

3.8.6. Immunohistochemical Detection of Cell Proliferation Ki-67 Antigen. Ki-67 immunoreactivity was concentrated in the nucleus of tumor cells. Its expression was determined as the percentage of cells positively stained by the antibody, which is considered a useful method for evaluating the proliferative activity of liver cancer cells.⁶⁴ In DEN-induced HCC, liver sections showed the highest expression level of Ki-67 ($\sim 93.76\%$) compared to the negative control group that displayed the lowest expression level (11.0%) (Figure 11B,C). The number of Ki-67 stained positive cells decreased significantly ($P < 0.05$) after treatment with free drug combination. Further reduction in Ki-67 expression level has been observed in groups treated with dual drug-loaded LF-NCs (62%) and GA/LF-NCs (20%). Furthermore, LA/LF-NCs-treated group revealed the lowest expression level, which was comparable to the negative control group, which proved their ability to suppress the tumor proliferation. The current results were in agreement with the findings of a previous study, where LA/folate dual-targeted casein micelles co-loaded with diosmin and berberine showed the lowest expression level of Ki-67 in DEN-induced HCC sections, which was also comparable to normal control group.¹⁷

4. CONCLUSIONS

In this study, lactoferrin shell-coated oily core nanocapsules (LF-NCs), glycyrrhetic acid (GA)-targeted and lactobionic acid (LA)-targeted LF-NCs, were fabricated for combined delivery of the hydrophobic drugs, SFB and QRC, in hepatocellular carcinoma treatment. All prepared NCs demonstrated sustained release of both drugs with excellent physical stability for 3 consecutive months with no remarkable changes in their particle size. Besides, a good hemocompatibility and serum stability have been observed for the prepared LF-based NCs. Furthermore, dual-targeted LA/LF-NCs exhibited enhanced cellular internalization and superior cytotoxicity against liver cancer cell line and HCC-induced mice. This superiority was manifested through activation of the apoptotic enzyme, caspase-3, and suppression of NF- κ B, TNF- α , and VEGF-1 expression levels compared to free combination in DEN-induced HCC mice ($P < 0.05$). LA/LF-NCs and GA/LF-NCs attenuated the liver toxicity induced by DEN in animal models and reduced tumor proliferation via downregulation of the proliferative tumor biomarker Ki-67. Combining the results obtained with the in vitro and in vivo analyses, we can conclude that ligand-tagged LF-NCs offered a precious nanosystem for combined SFB/QRC delivery with enhanced antitumor efficacy against HCC over the free drugs.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.9b10164.

Further information about the preparation and characterization of NCs; physicochemical characterization of dual drug-loaded NCs; in vitro cytotoxicity and cellular internalization; and quantification of angiogenesis and apoptosis by ELISA, PCR, histopathological analysis, and immunohistochemical analysis methodology (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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